

RAJESWARAN GAJENDRAN

Embedded Software Engineer

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Summary

Embedded Software Engineer with 4.5+ years of experience in embedded product development, Industrial communication protocols, debugging and testing. Experienced in Embedded C, AUTOSAR, CAN, Modbus, functional safety, MCAL and low-level debugging.

Skills

Programming Language and Drivers:

Embedded C, DMA, GPIO, Interrupt, PWM, MCAL Development, SPI, CAN, Low Level Driver Development, MISRA C

Architecture and Protocols: Semi-Autosar, RTOS, MODBUS, PROFIBUS, UDS (14229)

Safety and Security: Functional Safety (ISO26262), Fault Tree Analysis(FTA)

Tools: Trace32 debugger, CANalyzer, CANape, EB Tresos, Saleae Logic Analyzer, IAR, GIT, GitHub Copilot, Eclipse

Microcontroller: Aurix TC27X family, Renesas MCU, STM32F407, TLF35584(safety IC)

Experience

ZF Chennai, India
Software Engineer 03/2022 - 07/2025

- **Functional Safety:** Designed and implemented a fault-tolerant external watchdog monitoring system using Infineon TLF35584 for the AURIX TriCore TC27x controller.
- Achieved ASIL-D compliance (ISO 26262) by integrating voltage monitoring, windowed watchdog supervision, and fail-safe power cycling.
- **RTE integration** for a Semi-AUTOSAR-based platform using EB tresos Configurator, aligning task-event mapping and scheduling with system requirements.
- Executed Fault Tree Analysis (FTA) for wheel-speed and acceleration sensor systems using ISOGRAPH.
- Engineered **trap-handler mechanisms** for Infineon TriCore TC27x microcontrollers.
- **MCAL Development:** Optimized SPI driver configurations using EB tresos.
- CAN: Developed and analyzed CAN communication and validated signal frames using a 24 MHz logic analyzer.
- **ECUTEST:** Developed CAPL-based automation for ECU-TEST validation, reducing manual test effort by 50%.
- Created technical L1 documents based on customer requirements and used QAC for MISRA C static-code analysis.

GRUNDFOS Chennai, India
Product Development Engineer 07/2025 - Present

- Worked in an Agile software development environment, participating in sprint planning, daily stand-up meetings, sprint reviews, and retrospectives.
- Developed and debugged embedded software for the **CIM200 communication interface module based on a Renesas microcontroller**, supporting Modbus and fieldbus communication.
- Created a unit-testing framework for Ethernet/IP functionality to improve regression coverage, repeatability, and test automation.
- Leveraged GitHub Copilot to accelerate development of the Ethernet/IP unit-testing framework and improve software-development workflows.
- Added a Modbus configuration register and integrated it with the existing persistent handler and EEPROM storage, ensuring the AutoSync setting is retained across power cycles.
- Documented software development processes to improve cross-functional team collaboration and maintainability.

Projects

TEBS-Trailer Electronic Braking System

Worked on ZF's ASIL-D commercial vehicle braking system using Semi-AUTOSAR (CESAR) architecture, supporting real-time safety-critical performance.

Supported CAN-based truck-trailer communication for synchronized control of braking-related signals and validation.

iRSS-Roll Stability Support

Worked on rollover-risk detection and mitigation functionality for trailer applications. Supported brake-intervention logic and sensitivity calibration for different trailer configurations.

Projects

Fieldbus Team

Worked on product-to-fieldbus data integration, including scaling and mapping of product data for industrial communication protocols such as Modbus, PROFIBUS and EtherNet/IP. Supported integration of product parameters with fieldbus interfaces to ensure consistent data representation and communication across protocols.

Education

KLNCE

Chennai, India

Bachelor of Engineering in ECE

03/2012 - 03/2016

Cranes Varsit

Bangalore, India

Post Graduate Diploma in Embedded Systems

2021 - 2022